PROJECT REVIEW

INDIVIDUAL CONTRIBUTION REPORT
WEEK 1

COURSE CODE: 24ADI204
COURSE TITLE: DATA SCIENCE AND VISUALIZATION

Renewable Energy and Weather Analysis

Team Formation and Data Hunting
Pandas and NumPy

SUBMITTED BY

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INTRODUCTION

This report describes my contribution to the Week 1 activities of our team project, Renewable Energy and Weather Analysis. The focus was to understand the dataset, work with Pandas and NumPy, and prepare an initial numerical summary. I worked with the shared team notebook to review the data structure and interpret the first results.

DOMAIN AND PROBLEM STATEMENT

The project belongs to renewable energy and environmental data analytics. It aims to explore how weather conditions relate to solar energy, wind energy and total renewable energy generation. Week 1 establishes the dataset structure and highlights values that need closer inspection before further analysis.

WEEK 1 OBJECTIVES

1. Understand the dataset and its relevance to the project.
2. Load the data into a Pandas DataFrame and preview the first rows.
3. Check the number of rows, columns and available variables.
4. Select numerical columns using NumPy and calculate their means.
5. Record initial observations for the next stage of data inspection.

ABOUT THE DATASET

The supplied project report describes the dataset as a simulated daily log covering Coimbatore, Salem, Madurai, Tiruchirappalli and Erode. The saved notebook output confirms 510 records and 14 columns. The file is named `renewable_energy_weather.csv.xls` in the notebook and is read using `pd.read_csv()`.

Group	Variables
Identifiers	Date, Location
Weather measurements	Temperature_C, Humidity_pct, Wind_Speed_kmh, Solar_Irradiance_Wm2, Cloud_Cover_pct, Rainfall_mm, Air_Pressure_hPa
Energy measurements	Solar_Energy_kWh, Wind_Energy_kWh, Total_Renewable_Energy_kWh, Energy_Demand_kWh
Weather category	Weather_Condition

MY CONTRIBUTION

I contributed to the shared Week 1 work by reviewing the dataset variables, checking the loading and structure outputs, and interpreting the numerical means. I focused on identifying which columns describe weather and which describe energy generation or demand. The negative wind-speed and rainfall means provided clear reasons to inspect the raw data more carefully in the following week.

Tools used: Python, Google Colab, Pandas and NumPy. The report uses the code and saved outputs in the supplied notebook.

IMPLEMENTATION

The following code combines the dataset-loading step with the Week 1 summary. Imports used only in later weeks are omitted.

```
import pandas as pd
import numpy as np

df = pd.read_csv("renewable_energy_weather.csv.xls")
print("Dataset loaded successfully!")
print("Shape:", df.shape)
display(df.head())

print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
print("\nColumns:")
print(df.columns.tolist())
print("\nNumerical Mean:")
print(df.select_dtypes(include=np.number).mean())
```

RECORDED OUTPUT

The saved output shows: **Dataset loaded successfully!** Shape: **(510, 14)**; rows: **510**; columns: **14**. The numerical means below are rounded to four decimal places.

Numerical column	Raw mean
Temperature_C	13.1098
Humidity_pct	51.1839
Wind_Speed_kmh	-2.7133
Solar_Irradiance_Wm2	533.8919
Cloud_Cover_pct	23.6434
Rainfall_mm	-8.1030
Air_Pressure_hPa	990.6039
Solar_Energy_kWh	3.9031
Wind_Energy_kWh	116.9371
Total_Renewable_Energy_kWh	140.9655
Energy_Demand_kWh	197.5715

INTERPRETATION

The mean wind speed is -2.7133 km/h and the mean rainfall is -8.1030 mm. These are not physically meaningful averages for the recorded quantities. The initial preview also contains -999 entries. These findings show that the raw means need validation; they should not be treated as reliable weather or energy conclusions. Detailed missing-value checks and cleaning belong to later weeks.

CONTRIBUTION SUMMARY

Week 1 activity	Contribution and outcome
Dataset understanding	Reviewed the project domain and grouped the 14 variables into identifiers, weather measurements, energy measurements and a weather label.
Loading and preview	Worked with the shared Pandas loading step and reviewed the first five records to understand the layout.
Structure checks	Confirmed the recorded shape of 510 rows and 14 columns and reviewed the column list.
Numerical summary	Used the Week 1 numerical selection and mean calculation to understand the initial values.
Initial interpretation	Highlighted the negative wind-speed and rainfall means as issues for follow-up inspection.

CHALLENGES AND LEARNING

Mixed column types: Date, Location and Weather_Condition are not numerical measurements. Selecting columns with `select_dtypes(include=np.number)` allows means to be calculated only for numerical columns.

Unexpected file extension: The notebook uses a filename ending in .csv.xls but loads it with `read_csv()`. I learned to follow the actual data format and the working loading step rather than infer the format from the final extension alone.

Misleading averages: A program can calculate a mean successfully even when the source values contain invalid codes. I learned to check whether the results are reasonable for the quantity being measured.

WEEK 1 OUTCOME

The dataset was successfully loaded and its basic structure was documented in the shared notebook. My Week 1 contribution helped establish an understanding of the variables and the numerical summary. The unusual negative averages showed why the dataset needed further inspection before drawing conclusions about renewable energy.

NEXT STEPS

For Week 2, inspect the data types, first and last rows, descriptive statistics, missing values and duplicate rows. Investigate the -999 entries and confirm how they should be handled. Cleaning and visual analysis can then proceed in their respective later weeks.

SOURCE MATERIALS

1. Project_Review_V_Nidhish_25BAD067.pdf - supplied reference report, particularly the introduction and Week 1 section.
2. DSV_Week1_to_Week4_Renewable_Energy_Google_Colab.ipynb.txt - shared notebook, dataset-loading cell and Week 1 code with saved outputs.